

Trainings ▾

Preparatory Steps

with Electronically Actuated Injector

- Clean the area around the fuel fittings of debris and dirt.
- Remove the aftercooler assembly (to remove the main fuel drain manifold). Refer to Procedure 010-004 (</qs3/pubsys2/xml/en/procedures/28/28-010-004-tr.html>).

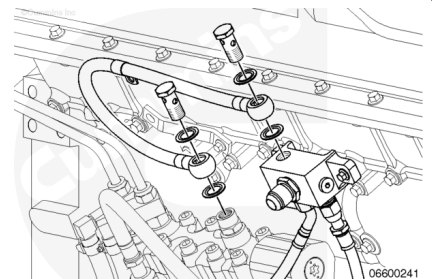


Remove

with Electronically Actuated Injector

⚠ WARNING ⚠

Use caution when draining fuel or replacing filter that the fuel is not spilled or drained into the bilge area. The fuel and filters must be disposed of in accordance with local environmental regulations.



Remove the banjo capscrews and the sealing washers securing the drain line between the mechanical dump valve and the drain manifold block.

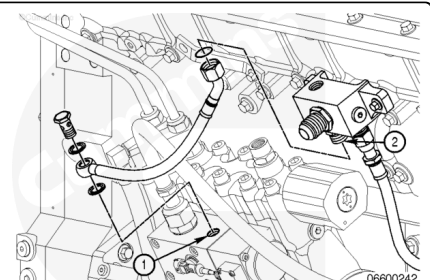
Discard the sealing washers.

Remove the banjo capscrews and the sealing washers securing the air bleed line that runs from the drain manifold block to the fuel pump.

Discard the sealing washers.

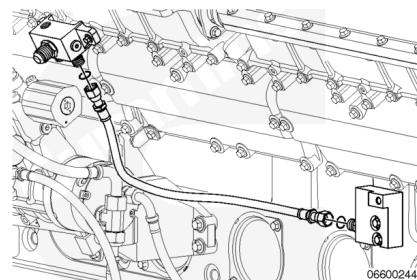
Remove the air bleed line from the drain manifold block.

Discard the o-ring.



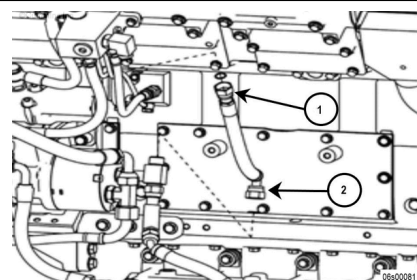
Remove the fuel drain line that connects the drain manifold to the fuel block manifold.

Discard the o-rings.

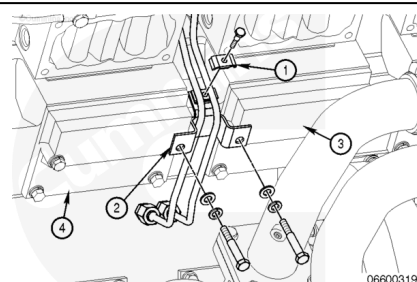


For engines fitted with a drain line between the fuel drain manifold (1), and the pump gerotor T fitting (2), remove the drain line and place into or cover with a plastic bag.

Discard used o-rings.

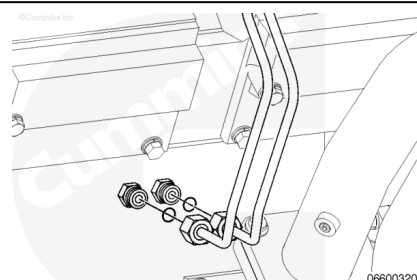


Remove the capscrew and upper part of the bracket (1) that holds the main fuel drain manifold to the bracket (2) that secures it on the second (3) and third (4) cam follower cover. Repeat for the opposite bank.



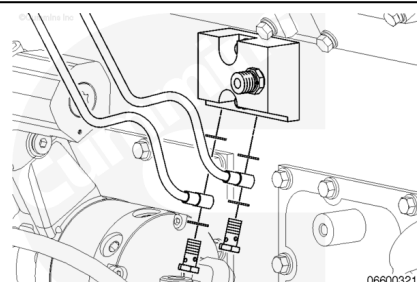
Disconnect the main fuel drain manifold on the right bank from the engine block by completely loosening the nuts.

Discard the o-rings.



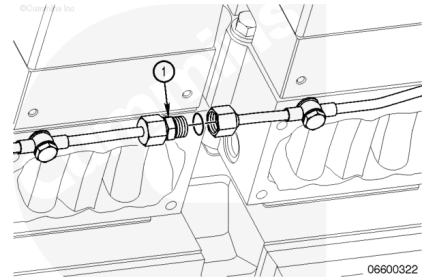
Disconnect the main fuel drain manifold on the left bank from the drain manifold block by removing both banjo capscrews and sealing washers.

Discard the sealing washers.



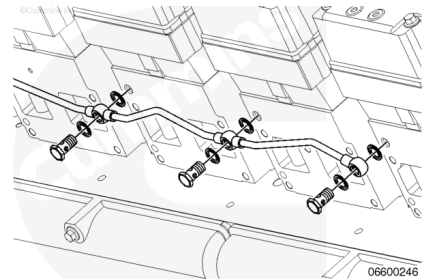
Completely loosen both segments of the main fuel drain manifold from the male union (1) on both banks.

Discard the o-rings.



Remove the remaining sixteen banjo capscrews and associated sealing washers that secure the main fuel drain manifold segments to the cylinder heads.

Discard the sealing washers.

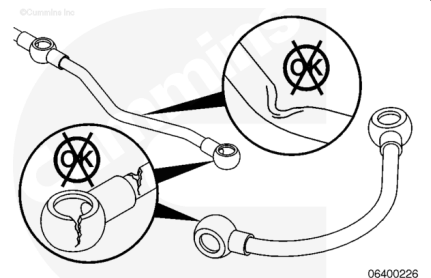


Inspect for Reuse with Electronically Actuated Injector

Inspect the drain lines for cracks and other visible damage. The drain lines must be replaced if they are cracked or damaged.

Inspect the banjo capscrews for visible damage.

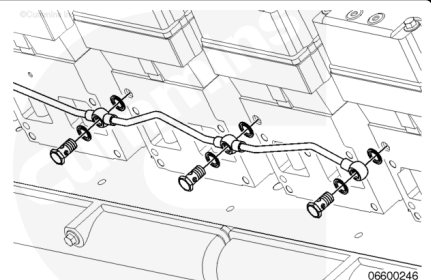
Replace as necessary.



Install with Electronically Actuated Injector

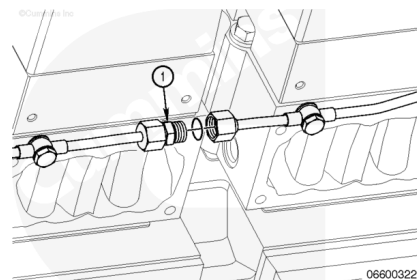
Install all sixteen segments of the fuel drain manifold using the sixteen banjo capscrews and new sealing washers.

Torque Value: 22 n•m [16 ft-lb]



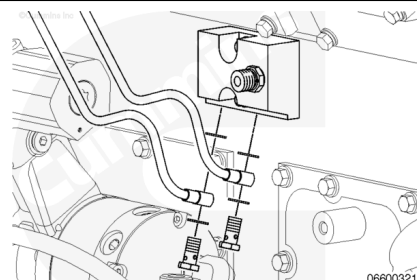
Attach both segments of the main fuel drain manifolds to the union (1) on both banks. Use new o-rings.

Torque Value: 25 n•m [18 ft-lb]



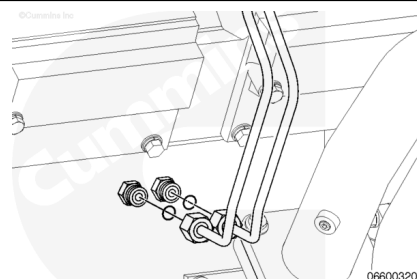
Install the main fuel drain manifold on the left bank to the drain manifold block, use both banjo capscrews and new sealing washers.

Torque Value: 22 n•m [16 ft-lb]



Install the main fuel drain manifold on the right bank on the engine block, use new o-ring.

Torque Value: 25 n•m [18 ft-lb]

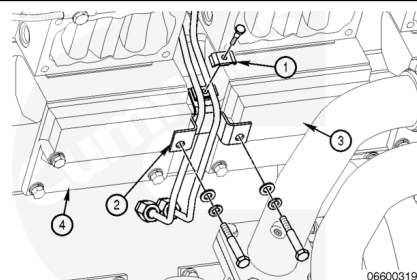


⚠ CAUTION ⚠

Do not over torque the capscrew as damage to the bracket will occur.

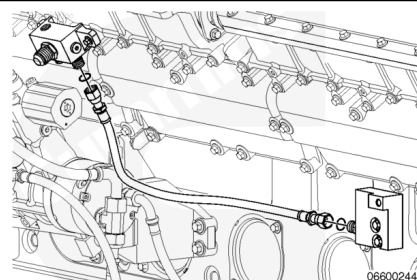
Install the capscrew and upper part of the bracket (1) that hold the main fuel drain line manifold to the bracket (2) that secures it on the second (3) and third (4) cam follower cover. Repeat on the opposite bank.

Torque Value: 5 n•m [44 in-lb]



Install the fuel drain line that connects the drain manifold block to the fuel block manifold. Use new o-rings at each end.

Torque Value: 45 n•m [33 ft-lb]

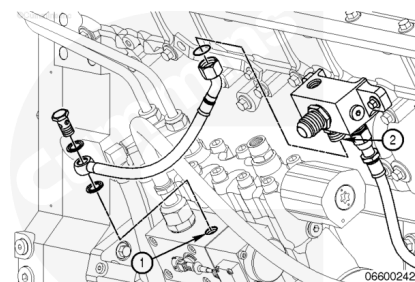


Install the banjo capscrew and new sealing washers securing the air bleed line that runs from the drain manifold block to the fuel pump.

Torque Value: 30 n•m [22 ft-lb]

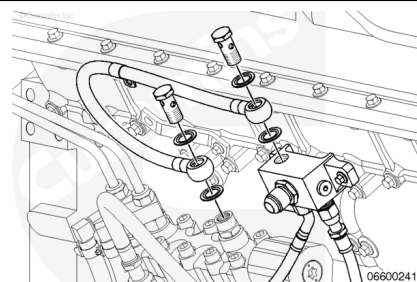
Install the air bleed line from the drain manifold block. Use a new o-ring.

Torque Value: 45 n•m [33 ft-lb]



Install the banjo capscrews and new sealing washers securing the drain line between the mechanical dump valve and the drain manifold block.

Torque Value: 48 n•m [35 ft-lb]



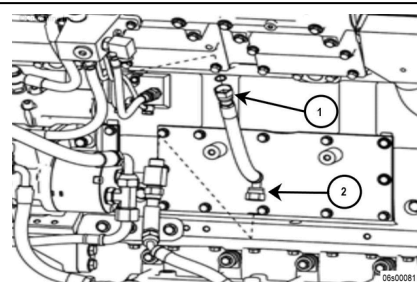
Remove the fuel drain line from the protective plastic bag. Replace the o-ring.

Install the line between the fuel manifold drain elbow (1), and the upper connect of the gerotor rear t-piece (2).

Torque Value:

1. 80 n•m [59 ft-lb]

2. 80 n•m [59 ft-lb]



Finishing Steps

with Electronically Actuated Injector

- Install the aftercooler assembly (if main fuel drain manifold was removed). Refer to Procedure 010-004 (/qs3/pubsys2/xml/en/procedures/28/28-010-004-tr.html).
- Operate the engine and check for leaks.

